Sharp Identification Regions in General Selection Models with (Un)ordered Treatments and Discrete Instruments

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This paper: Instrumental Variables (IV) with Multiple Treatments

Setting:

- Finitely-supported treatment: $\mathcal{D}$ is finite, ordered or unordered
- Finitely-supported instrument: $\mathcal{Z}$ is finite, common in empirical work
- Y may be continuous: $\mathcal{Y}$ compact convex subset of $\mathbb{R}^n$, or finite
- Randomly assigned instruments, not just mean independent

Goal: Unified (partial) identification approach that allows flexible choice of target parameter and treatment selection (or choice) model

Scope: Models that impose assumptions on selection, but not outcome

Model

Observables (Y, D, Z), unobservables $(\boldsymbol{V}, \varepsilon_Y)$, functions g_D and g_Y:

$$\text{Treatment Selection Equation:} \quad D = g_D(Z, \boldsymbol{V}) \ ,$$
$$\text{Outcome Equation:} \quad Y = g_Y(D, \boldsymbol{V}, \varepsilon_Y) \ ,$$
$$\text{Independence Condition:} \quad Z, \boldsymbol{V}, \varepsilon_Y \text{ are mutually independent.}$$

So far, **formally equivalent** to usual potential outcomes IV model, define potential outcomes and treatments:

$$Y_d := g_Y(d, \boldsymbol{V}, \varepsilon_Y) \ ,$$
$$D_z := g_D(z, \boldsymbol{V}) \ .$$

This paper:

Models that impose assumptions on selection, but not outcome

→ **Allow:** assumptions on treatment selection equation,
i.e. restrictions on g_D and/or on $\boldsymbol{V}$

→ **Avoid:** assumptions on outcome equation

Selection Models and Response-Types

Define an individual's ***response-type***, denoted $\boldsymbol{T}$, as the random vector

$$\boldsymbol{T}(\boldsymbol{V}) = \boldsymbol{T} := (D_z : z \in \mathcal{Z}) = (g_D(z, \boldsymbol{V}) : z \in \mathcal{Z})$$

$\rightarrow$ Conditional on $\boldsymbol{T}$, D is randomly assigned (determined by Z)

$\rightarrow$ Selection models restrict the set of admissible distributions for $\boldsymbol{T}$

Heckman and Pinto (2018): discrete IV identification problem is finite mixture problem involving response-type conditional distributions

$\rightarrow$ **This paper:** attack mixture problem directly, develop new results on (partial) identification of components in such mixture models

Overview of Results

For any selection model, paper provides **tractable** characterizations of

1. their empirical content and (sharp) testable implications,
2. the identified set for a large class of target parameters, including mean counterfactuals such as counterfactual choice probabilities, mean outcomes, and mean treatment effects,
3. the identified set for (joint) potential outcome distributions.

$\rightarrow$ Results are **sharp under random assignment** of Z

The proposed approach can be used to derive:

- **Analytic bounds** on target parameters $\rightarrow$ not “black box”
- Inclusion tests for inference on target parameters
  - $\rightarrow$ can be inverted to construct confidence regions
- Specification tests for treatment selection models

For “large” problems where analytic methods may be impractical, propose computationally feasible algorithm.

Example: Moving to Opportunity (MTO) Experiment

Let D denote neighbourhood choice. Here, individuals choose from three different types of neighbourhoods: "high" poverty (H), "medium" poverty (M), and "low" poverty (L).

Let $Z = (Z_M, Z_L)$ be the randomly assigned instrument. Individuals living in high poverty neighbourhoods were randomly offered rent vouchers:

$(Z_M, Z_L) = (0, 0)$: No vouchers

$(Z_M, Z_L) = (0, 1)$: Vouchers for L neighbourhoods

$(Z_M, Z_L) = (1, 1)$: Vouchers for M and L neighbourhoods

→ Therefore, $D \in \{H, M, L\}$ and $Z \in \{(0, 0), (0, 1), (1, 1)\}$.

→ Response-type: $\boldsymbol{T} = \left(D_{(0,0)}, D_{(0,1)}, D_{(1,1)}\right)$.

Example: Moving to Opportunity (MTO) Experiment

Suppose we impose "Unordered Partial Monotonicity" (Mountjoy, 2022):

$\rightarrow$ Z_M incentivizes $D = M$ and Z_L incentivizes $D = L$

$\rightarrow$ Changes in Z don't effect choices among irrelevant alternatives (e.g., $Z : (0,0) \rightarrow (0,1)$ cannot cause $D : H \rightarrow M$)

Shifting $Z : (1,1) \rightarrow (0,1)$ causes two types of moves:

- Individuals with $\boldsymbol{T} \in \{(H, L, M), (M, L, M), (L, L, M)\}$: $M \rightarrow L$
- Individuals with $\boldsymbol{T} = (H, H, M)$: $M \rightarrow H$

The following weighted average can be (point) identified

$$\omega_{M \rightarrow L} E\left[Y_L - Y_M \mid D_{(0,1)} = L, D_{(1,1)} = M\right] \\ + \omega_{M \rightarrow H} E\left[Y_H - Y_M \mid D_{(0,1)} = H, D_{(1,1)} = M\right],$$

but the two component effects are only partially identified.

Other Examples of Potential Applications

- Multi-dimensional instrument(s) or treatment(s)
  - Multi-dimensional treatment: disentangling different dimensions
  - Mediation: want to separate direct, indirect, and total effects
  - More generally: settings with complex incentive schemes, or complicated relationship between Z and D
- Ordered Treatments: e.g., Impact of Incarceration, Schooling etc.
- Nonparametric Discrete Choice Models
  - MTO, Head Start, College Choice...
- Other settings with multi-dimensional heterogeneity
  - Survey/RCT Non-Response Bias (e.g., Dutz et al., 2022)
- Exclusion Restriction Violations and/or Sample Selection Problems
  - See, e.g., Kroft, Mourifié, and Vayalinkal (2024)
- Sensitivity analysis for point-identified models
  - In the spirit of Noack (2021)

Parameters of Interest

Today: For simplicity:

- assume scalar continuous Y, supported on an interval $\mathcal{Y} \subseteq \mathbb{R}$
- focus on models that simply eliminate some response-types
  - $\rightarrow$ i.e. models that restrict

$$\mathcal{T} := \operatorname{supp}(\boldsymbol{T}) = \left\{\boldsymbol{\tau}_1, \ldots, \boldsymbol{\tau}_{N_{\boldsymbol{T}}}\right\}$$

  (basic example: "monotonicity" = "no defiers")

- focus on (weighted averages of) parameters of the form

$$E\left[Y_d - Y_{d'} \mid \boldsymbol{T} = \boldsymbol{\tau}\right],$$

  - $\rightarrow$ examples include ATE, ATT, ATUT, (generalized) LATE, PRTE etc.

Related IV Literature

Point Identification: Angrist and Imbens (1995)... Heckman and Vytlacil (1999, 2001, 2007a,b)... Kirkebøen et al. (2016); Heckman and Pinto (2018); Lee and Salanié (2018); Mountjoy (2022)...

Partial Identification: Manski (1989, 1990); Balke and Pearl (1997)... Beresteanu et al. (2012)... Mogstad et al. (2018); Torgovitsky (2019); Kitagawa (2021); Kamat et al. (2023); Marx (2024)...

→ **This paper:** sharp under full IV independence for any selection model; allows multivalued treatments and continuous outcomes; tractable, but not "black box"

→ **This paper cannot:** incorporate general assumptions on outcomes or continuous instruments

Specification Testing : Kitagawa (2015); Mourifié and Wan (2017); Kédagni and Mourifié (2020); Sun (2023)...

→ **This paper:** sharp test for general selection models with discrete IV

IV Mixture Model: Basic Idea

For any $d \in \mathcal{D}$ and each $z \in \mathcal{Z}$,

We observe Y for those assigned $Z = z$ and took treatment d

We want to learn about Y_d for those of type $\boldsymbol{T} = \boldsymbol{\tau}$, for some $\boldsymbol{\tau} \in \mathcal{T}$

Key: Individuals who take treatment $D = d$ when $Z = z$ consists of the mixture of individuals of types $\boldsymbol{T} = \boldsymbol{\tau}$ such that $\boldsymbol{\tau}(z) = d$.

$\rightarrow$ Observed outcome distributions are mixture distributions:

$$\sum_{\boldsymbol{\tau} \in \mathcal{T}} \underbrace{\mathbb{1}\{\boldsymbol{\tau}(z) = d\}}_{\text{known}} \underbrace{P(\boldsymbol{T} = \boldsymbol{\tau}) \, P(Y_d \leq y \mid \boldsymbol{T} = \boldsymbol{\tau})}_{\text{unobserved}}$$

$$= \underbrace{P(D = d \mid Z = z) \, P(Y \leq y \mid D = d, Z = z)}_{\text{observed}},$$

$\rightarrow$ Allowing Z to vary, get linear system with known coefficients

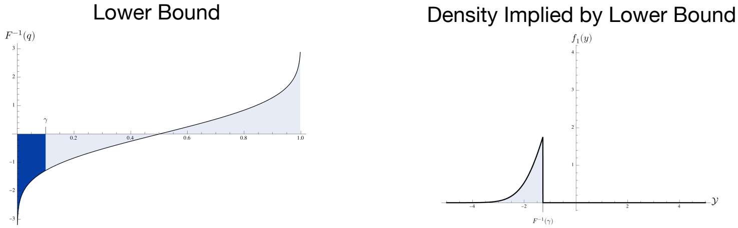
"Trimming" Bounds of Horowitz and Manski (1995)

Suppose all we observe is the following mixture distribution, with CDF F,

$$F(y) = \gamma F_1(y) + (1-\gamma) F_2(y) \ ,$$

where $\gamma \in (0,1)$ is known, but F_1 and F_2 are unknown CDFs. Let $Y_1 \sim F_1$.

If $\gamma = 0.1$, sharp lower bound for $E[Y_1]$ is constructed as follows:



(a) Lower bound is area of shaded region

(b) Density of F_1

Figure 1: Sharp lower bound for $E[Y_1]$, with $\gamma = 0.1$

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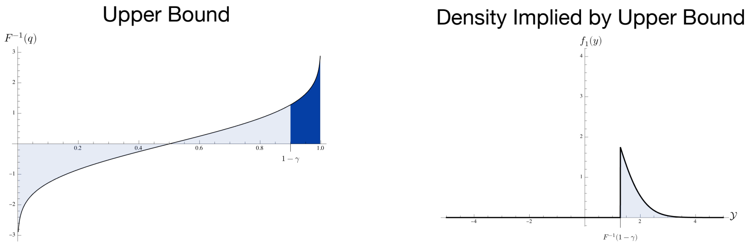
If $\gamma = 0.1$, sharp **upper** bound for $E[Y_1]$ is constructed as follows:



(a) Upper bound is area of shaded region

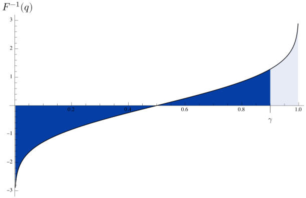
(b) Density of F_1

Figure 1: Sharp upper bound for $E[Y_1]$, with $\gamma = 0.1$

“Trimming” Bounds of Horowitz and Manski (1995)

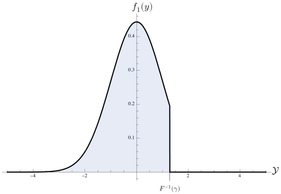
If instead we have $\gamma = 0.9$, then lower bound is

Lower Bound



(a) Lower bound is area of shaded region

Density Implied by Lower Bound

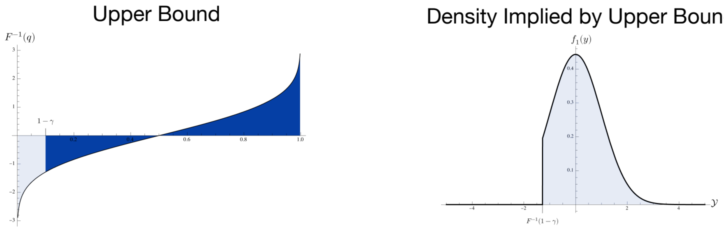


(b) Density of F_1

Figure 2: Sharp lower bound for $E[Y_1]$, with $\gamma = 0.9$

“Trimming” Bounds of Horowitz and Manski (1995)

and upper bound is



(a) Upper bound is area of shaded region

(b) Density of F_1

Figure 2: Sharp upper bound for $E\left[Y_1\right]$, with $\gamma = 0.9$

Horowitz and Manski (1995) Bounds: Takeaway

The observed mixture distribution, written in terms of densities, is

$$f\left(y\right)=\gamma f_1\left(y\right)+\left(1-\gamma\right)f_2\left(y\right)\ ,$$

where f_1 is the density of F_1, and f_2 is the density of F_2.

This implies two restrictions on the function $\gamma f_1\left(\cdot\right)$:

→ point-wise restriction: $0\leq\gamma f_1\left(y\right)\leq f\left(y\right)$, for all y

→ mass restriction: $\int_{-\infty}^{\infty}f_1\left(y\right)d\mu\left(y\right)=1 \implies \int_{-\infty}^{\infty}\gamma f_1\left(y\right)d\mu\left(y\right)=\gamma$

Note that $E\left[Y_1\right]:=\dfrac{1}{\gamma}\displaystyle\int_{-\infty}^{\infty}y\gamma f_1\left(y\right)\ d\mu\left(y\right)$

→ optimizing the right hand side over all functions $\gamma f_1\left(\cdot\right)$ that satisfy the two conditions above results in the Horowitz-Manski bounds

Example 1: Head Start Impact Study

Let D denote preschool choice. Families choose between no preschool (n), competing preschools (c), and Head Start (h) for their child.

The Head Start Impact Study randomized applicants into two groups:

$Z = 0$: Not offered admission to Head Start

$Z = 1$: Offered admission to Head Start.

→ Therefore, $D \in \{n, c, h\}$ and $Z \in \{0, 1\}$.

→ Response-type: $\boldsymbol{T} = (D_0, D_1)$.

Example 1 (Head Start): Partial Identification

Assume that the following assumption holds for everyone

- either $D_1 = D_0$ or $\mathbb{1}\{D_1 = h\} > \mathbb{1}\{D_0 = h\}$

Shifting $Z : 0 \to 1$ causes two types of enrolment changes:

- Individuals with $\boldsymbol{T} = (n, h)$: $n \to h$
- Individuals with $\boldsymbol{T} = (c, h)$: $c \to h$

The following weighted average can be (point) identified

$$\omega_{n \to h} E\left[Y_h - Y_n \mid D_0 = n, D_1 = h\right] \\ + \omega_{c \to h} E\left[Y_h - Y_c \mid D_0 = c, D_1 = h\right] \ ,$$

but the two component effects are only partially identified.

Example 1 (Head Start): "h"-response-type

Consider the effect $E\left[Y_h - Y_c \mid \boldsymbol{T} = (ch)\right]$

$\rightarrow$ $P\left[\boldsymbol{T} = (ch)\right]$ and $E\left[Y_c \mid \boldsymbol{T} = (ch)\right]$ are point-identified

$\rightarrow$ Therefore, it suffices to find the bounds for

$$\begin{aligned} P\left[\boldsymbol{T} = (ch)\right] E\left[Y_h \mid \boldsymbol{T} = (ch)\right] &= E\left[Y_h \mathbb{1}\left\{\boldsymbol{T} = (ch)\right\}\right] \\ &= \int_{\mathcal{Y}} P\left[\boldsymbol{T} = (ch)\right] f_{Y_h \mid \boldsymbol{T} = (ch)}(y)\, d\mu(y) \end{aligned}$$

There are two types such that $D_0 \neq h$ and $D_1 = h$:

$$\{(ch), (nh)\}$$

Example 1 (Head Start): Pointwise Restriction

Define

$$\boldsymbol{\xi}_{Y_h|\boldsymbol{T}=(ch)}(y) := P[\boldsymbol{T}=(ch)]\, f_{Y_h|\boldsymbol{T}=(ch)}(y),$$

and $$\boldsymbol{\xi}_{Y_h|\boldsymbol{T}=(nh)}(y) := P[\boldsymbol{T}=(nh)]\, f_{Y_h|\boldsymbol{T}=(nh)}(y).$$

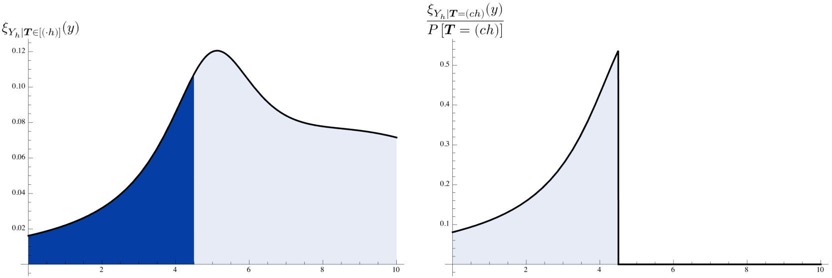
Now, note that

$$\begin{aligned} & P(D=h|Z=1)\, f_{Y|D=h,Z=1}(y) - P(D=h|Z=0)\, f_{Y|D=h,Z=0}(y) \\ = \; & \boldsymbol{\xi}_{Y_h|\boldsymbol{T}=(ch)}(y) + \boldsymbol{\xi}_{Y_h|\boldsymbol{T}=(nh)}(y) \end{aligned}$$

Defining $\boldsymbol{\xi}_{Y_h|\boldsymbol{T}\in[(\cdot h)]} := \boldsymbol{\xi}_{Y_h|\boldsymbol{T}=(ch)}(y) + \boldsymbol{\xi}_{Y_h|\boldsymbol{T}=(nh)}(y)$, we can "trim" the identified function $\boldsymbol{\xi}_{Y_h|\boldsymbol{T}=[(\cdot h)]}$ to get sharp bounds on $E[Y_h|\boldsymbol{T}=(ch)]$.

Example 1 (Head Start): Sharp Lower Bound

Suppose $P\left[\boldsymbol{T} = (ch)\right] = 0.2$.



(a) Dark blue region has area 0.2.

(b) Density $\frac{\xi_{Y_h|\boldsymbol{T}=(ch)}}{P[\boldsymbol{T}=(ch)]} = f_{Y_h|\boldsymbol{T}=(ch)}$ implied by sharp lower bound

Figure 3: (Simulated) Construction of sharp lower bound for $E\left[Y_h|\boldsymbol{T} = (ch)\right]$

Example 2 (MTO): "M"-response-type

Recall: $D \in \{H, M, L\}$ and $Z = (Z_M, Z_L) \in \{(0,0), (0,1), (1,1)\}$.
Response-type: $\boldsymbol{T} = \left(D_{(0,0)} D_{(0,1)} D_{(1,1)}\right)$.

Consider the effect $E\left[Y_H - Y_M \mid \boldsymbol{T} = (HHM)\right]$

→ $P\left[\boldsymbol{T} = (HHM)\right]$ and $E\left[Y_H \mid \boldsymbol{T} = (HHM)\right]$ are point-identified

→ Therefore, it suffices to find the bounds for

$$P\left[\boldsymbol{T} = (HHM)\right] E\left[Y_M \mid \boldsymbol{T} = (HHM)\right] = \int_{\mathcal{Y}} P\left[\boldsymbol{T} = (HHM)\right] f_{Y_M \mid \boldsymbol{T} = (HHM)}(y)\, d\mu(y)$$
$$=: \int_{\mathcal{Y}} \boldsymbol{\xi}_{Y_M \mid \boldsymbol{T} = (HHM)}(y)\, d\mu(y)$$

There are three types such that $D_{(0,0)} \neq M, D_{(0,1)} \neq M$ and $D_{(1,1)} = M$:

$$\{(HHM), (HLM), (LLM)\}$$

Example 2 (MTO): Pointwise Restrictions

In this case, the aggregate density

$$\boldsymbol{\xi}_{Y_M|\boldsymbol{T}\in[(\cdot\cdot M)]}(y) := \boldsymbol{\xi}_{Y_M|\boldsymbol{T}=(HHM)}(y) + \boldsymbol{\xi}_{Y_M|\boldsymbol{T}=(HLM)}(y) + \boldsymbol{\xi}_{Y_M|\boldsymbol{T}=(LLM)}(y)$$

is not point-identified, but it can be bounded.

Here,

$$\begin{aligned} & P[D=M|Z=(1,1)]\, f_{Y|D=M,Z=(1,1)}(y) \\ = & \boldsymbol{\xi}_{Y_M|\boldsymbol{T}\in[(\cdot\cdot M)]}(y) + \boldsymbol{\xi}_{Y_M|\boldsymbol{T}=(MMM)}(y) + \boldsymbol{\xi}_{Y_M|\boldsymbol{T}=(MLM)}(y) \ . \end{aligned}$$

First, we have that

$$\begin{aligned} & P[D=M|Z=(1,1)]\, f_{Y|D=M,Z=(1,1)} - P[D=M|Z=(0,1)]\, f_{Y|D=M,Z=(0,1)} \\ = & \boldsymbol{\xi}_{Y_M|\boldsymbol{T}\in[(\cdot\cdot M)]} + \boldsymbol{\xi}_{Y_M|\boldsymbol{T}=(MLM)} \\ \geq & \boldsymbol{\xi}_{Y_M|\boldsymbol{T}\in[(\cdot\cdot M)]} \end{aligned}$$

Define $\overline{f}^*(y)$ to be this bound.

Example 2 (MTO): Pointwise Restrictions

In this case, the aggregate density

$$\boldsymbol{\xi}_{Y_M|\boldsymbol{T}\in[(\cdot\cdot M)]}(y) := \boldsymbol{\xi}_{Y_M|\boldsymbol{T}=(HHM)}(y) + \boldsymbol{\xi}_{Y_M|\boldsymbol{T}=(HLM)}(y) + \boldsymbol{\xi}_{Y_M|\boldsymbol{T}=(LLM)}(y)$$

is not point-identified, but it can be bounded.

Here,

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Next, we have that

$$\begin{aligned} & P[D=M|Z=(1,1)]\, f_{Y|D=M,Z=(1,1)} - P[D=M|Z=(0,0)]\, f_{Y|D=M,Z=(0,0)} \\ = & \boldsymbol{\xi}_{Y_M|\boldsymbol{T}\in[(\cdot\cdot M)]} - \boldsymbol{\xi}_{Y_M|\boldsymbol{T}=(MLL)} \\ \leq & \boldsymbol{\xi}_{Y_M|\boldsymbol{T}\in[(\cdot\cdot M)]} \end{aligned}$$

Define $\underline{f}^*(y)$ to be the pointwise maximum of 0 and this bound.

Example 2 (MTO): Simple Bounds

So, we have the pointwise bounds $\underline{f}^*(y) \leq \boldsymbol{\xi}_{Y_M|\boldsymbol{T}\in[(\cdot\cdot M)]}(y) \leq \overline{f}^*(y)$:

→ valid bound on $P[\boldsymbol{T} = (HHM)]\, E[Y_M|\boldsymbol{T} = (HHM)]$ can be obtained by trimming $\overline{f}^*(y)$

→ however, this bound is not generally sharp

This is because the resulting bound may imply a value of

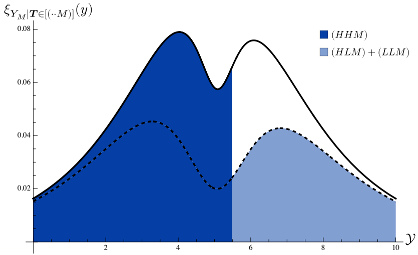
$$P[\boldsymbol{T} \in [(\cdot\cdot M)]] := P[\boldsymbol{T} = (HHM)] + P[\boldsymbol{T} = (HLM)] + P[\boldsymbol{T} = (LLM)]$$

that may not be feasible.

Example 2 (MTO): Non-Sharpness of Simple Bounds

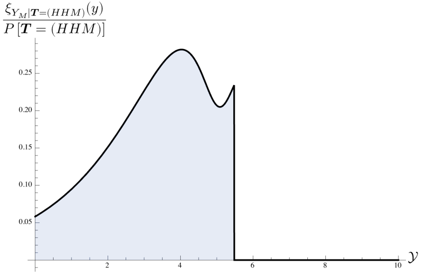
Suppose the sharp upper bound on $P\left[\boldsymbol{T} \in [(\cdot\cdot M)]\right]$ is 0.36, and $P\left[\boldsymbol{T} = (HHM)\right] = 0.28$.

Construction of Simple Lower Bound



dark blue area $= 0.28$
dark blue area + light blue area $= \mathbf{0.423}$

Density $f_{Y_M|\boldsymbol{T}=(HHM)}$ Implied by Simple Lower Bound



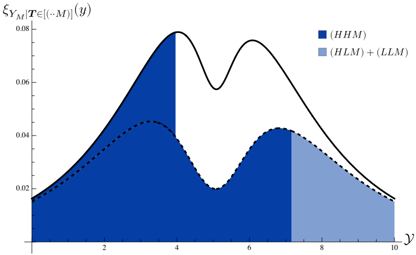
This density implies the simple lower bound of 3.289

Figure 4: (Simulated) Construction of lower bounds for $E\left[Y_M|\boldsymbol{T} = (HHM)\right]$

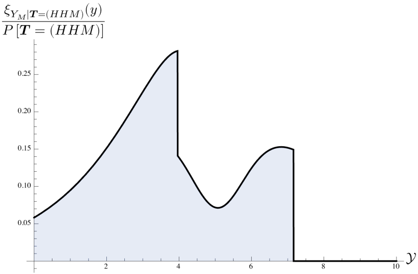
Example 2 (MTO): Non-Sharpness of Simple Bounds

Suppose the sharp upper bound on $P\left[\boldsymbol{T} \in [(\cdot\cdot M)]\right]$ is 0.36, and $P\left[\boldsymbol{T} = (HHM)\right] = 0.28$.

Construction of Sharp Lower Bound



Density $f_{Y_M|\boldsymbol{T}=(HHM)}$ Implied by Sharp Lower Bound



dark blue area $= 0.28$
dark blue area + light blue area $= \mathbf{0.36}$

This density implies the sharp lower bound of 3.665

Figure 4: (Simulated) Construction of lower bounds for $E\left[Y_M|\boldsymbol{T} = (HHM)\right]$

Example 3: Multi-dimensional Treatment

Consider a setting with a primary treatment, $P \in \{0,1\}$, and supplementary treatment, $S \in \{0,1\}$, and define $D = (P, S)$.

Some individuals are randomly offered incentives to participate in treatment. Let $Z = (Z_P, Z_S)$ where

- $Z_P \in \{0,1\}$ represents an incentive to participate in P
- $Z_S \in \{0,1\}$ represents an additional incentive to participate in S

Suppose individuals are

- only offered supplementary incentive if offered primary incentive
- only allowed supplementary treatment if they take primary treatment

$\rightarrow$ Therefore, $D \in \{(0,0),(1,0),(1,1)\}$ and $Z \in \{(0,0),(1,0),(1,1)\}$.

$\rightarrow$ Response-type: $\boldsymbol{T} = \left(D_{(0,0)}, D_{(1,0)}, D_{(1,1)}\right)$.

Multi-dimensional Treatment Example

Assume that the following holds for all individuals:

1. $Z_P : 0 \to 1$ (weakly) increases $\mathbb{1}\{D \in \{(1,0),(1,1)\}\}$ and $\mathbb{1}\{S = 1\}$,
2. $Z_S : 0 \to 1$ (weakly) increases $\mathbb{1}\{D = (1,1)\}$ and $\mathbb{1}\{P = 1\}$.

Many potential parameters of interest...

For example, consider the decomposition

$$\underbrace{E\left[Y_{(1,1)} - Y_{(0,0)} \mid \boldsymbol{T} = ((0,0),(1,0),(1,1))\right]}_{\text{Total Effect}}$$

$$= \underbrace{E\left[Y_{(1,1)} - Y_{(1,0)} \mid \boldsymbol{T} = ((0,0),(1,0),(1,1))\right]}_{\text{Supplementary Effect}}$$

$$+ \underbrace{E\left[Y_{(1,0)} - Y_{(0,0)} \mid \boldsymbol{T} = ((0,0),(1,0),(1,1))\right]}_{\text{Primary Effect}}$$

$\to$ **All components are only partially identified!**

Multi-dim. Treatment example: Singleton "$(1,0)$"-response-type

Let's focus on the mean outcome $E\left[Y_{(1,0)} \,\middle|\, \boldsymbol{T} = (00,10,11)\right]$.

$\rightarrow$ there are no other response-types such that $D_{(0,0)} \neq (1,0), D_{(1,1)} \neq (1,0)$ and $D_{(1,0)} = (1,0)$

$\rightarrow$ $\boldsymbol{\xi}_{Y_{(1,0)}|\boldsymbol{T}=(00,10,11)}$ is not point-identified, but can be bounded

Define $\mathbf{pf}_{d|Z=z}(y) := P(D = d|Z = z)\, f_{Y|D=d,Z=z}(y)$.

Proceeding as before, we have the bounds

$$\begin{aligned}\underline{f}^*(y) :=\ & \max\left\{0, \mathbf{pf}_{(1,0)|Z=(1,0)}(y) - \mathbf{pf}_{(1,0)|Z=(1,1)}(y) - \mathbf{pf}_{(1,0)|Z=(0,0)}(y)\right\} \\ & \leq \boldsymbol{\xi}_{Y_{(1,0)}|\boldsymbol{T}=(00,10,11)}(y) \leq \\ & \quad \mathbf{pf}_{(1,0)|Z=(1,0)}(y) - \mathbf{pf}_{(1,0)|Z=(1,1)}(y) =: \overline{f}^*(y)\ ,\end{aligned}$$

Multi-dim. Treatment example: Trimming Failure

We have $\underline{f}^{*}(y) \leq \boldsymbol{\xi}_{Y_{(1,0)}|\boldsymbol{T}=(00,10,11)}(y) \leq \overline{f}^{*}(y)$

For simplicity, assume that $P[\boldsymbol{T}=(00,10,11)]$ is identified, and define

$$\overline{p} := P[\boldsymbol{T}=(00,10,11)] - \int_{\mathcal{Y}} \underline{f}^{*}(y)\ d\mu(y)\ .$$

Simple bounds can be constructed as follows:

(i) Trim $\overline{f}^{*} - \underline{f}^{*}$ up to mass $\overline{p}$

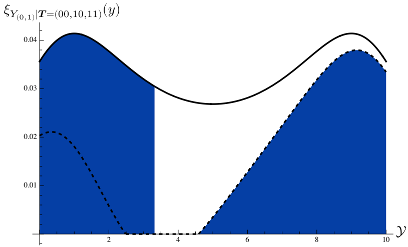
(ii) Add this to the density implied by lower bound, $\underline{f}^{*}(y)$

$\rightarrow$ The resulting bounds are valid, but **still not generally sharp**.

Multi-dim. Treatment example: Tighter Bounds due to Pointwise Dependence

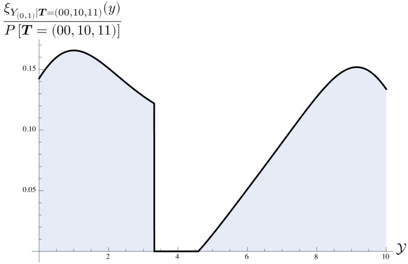
Suppose $P\left[\boldsymbol{T} = (00, 10, 11)\right] = 0.25$.

Construction of Simple Lower Bound



dark blue area $= 0.25$

Density $f_{Y_{(1,0)}|\boldsymbol{T}=(00,10,11)}$ Implied by Simple Lower Bound



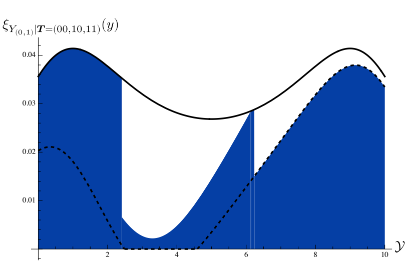
This density implies the simple lower bound of 4.865

Figure 5: Construction of lower bounds for $E\left[Y_{(1,0)}|\boldsymbol{T} = (00, 10, 11)\right]$

Multi-dim. Treatment example:
Tighter Bounds due to Pointwise Dependence

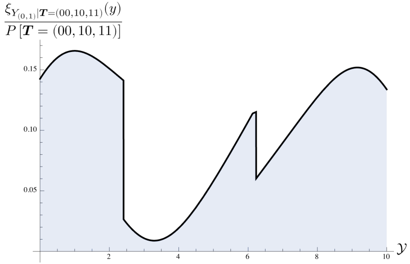
Suppose $P\left[\boldsymbol{T} = (00, 10, 11)\right] = 0.25$.

Construction of Sharp Lower Bound



dark blue area $= 0.25$

Density $f_{Y_{(1,0)}|\boldsymbol{T}=(00,10,11)}$ Implied by Sharp Lower Bound



This density implies the sharp lower bound of 5.106

Figure 5: Construction of lower bounds for $E\left[Y_{(1,0)}|\boldsymbol{T} = (00, 10, 11)\right]$

IV Mixture Model: Notation

Enumerate instrument values as $\mathcal{Z} := \{1, \dots, N_Z\}$. For any random variables X_1, X_2, let $f_{X_1|X_2}$ be the density of $X_1|X_2$.

For $d \in \mathcal{D}$ and $y \in \mathcal{Y}$, define $\mathbf{pf}_d$ as the **observed** vector of weighted densities

$$\mathbf{pf}_d(y) := \begin{bmatrix} P(D = d|Z = 1)\, f_{Y|D=d,Z=1}(y) \\ \vdots \\ P(D = d|Z = N_Z)\, f_{Y|D=d,Z=N_Z}(y) \end{bmatrix}$$

Define $\boldsymbol{\pi}$ as the **unobserved** vector of response-type probabilities,

$$\boldsymbol{\pi} := \begin{bmatrix} P(\boldsymbol{T} = \boldsymbol{\tau}_1) \\ \vdots \\ P\left(\boldsymbol{T} = \boldsymbol{\tau}_{N_{\boldsymbol{T}}}\right) \end{bmatrix}$$

and define $\boldsymbol{\xi}_d$ as the **unobserved** vector of weighted response-type densities

$$\boldsymbol{\xi}_d(y) := \begin{bmatrix} P(\boldsymbol{T} = \boldsymbol{\tau}_1)\, f_{Y_d|\boldsymbol{T}=\boldsymbol{\tau}_0}(y) \\ \vdots \\ P\left(\boldsymbol{T} = \boldsymbol{\tau}_{N_{\boldsymbol{T}}}\right) f_{Y_d|\boldsymbol{T}=\boldsymbol{\tau}_{N_{\boldsymbol{T}}}}(y) \end{bmatrix}$$

IV Mixture Model: Notation

Finally, define the d-**response-matrix** as

$$\mathbf{R}_d := \begin{bmatrix} \mathbb{1}\{\boldsymbol{\tau}_1(1) = d\} & \dots & \mathbb{1}\left\{\boldsymbol{\tau}_{N_T}(1) = d\right\} \\ \vdots & \ddots & \vdots \\ \mathbb{1}\{\boldsymbol{\tau}_0(N_Z) = d\} & \dots & \mathbb{1}\left\{\boldsymbol{\tau}_{N_T}(N_Z) = d\right\} \end{bmatrix},$$

which is the **known** binary matrix that indicates when each response-type takes treatment d (as in Heckman and Pinto, 2018).

Basic Example ($Z \in \{0,1\}$, $D \in \{0,1\}$, no restrictions on selection) **:** In this case, the response types are always-takers (AT), compliers (C), defiers (D), never-takers (NT) , leading to

$$\mathbf{R}_0 = \begin{array}{c} \begin{array}{cccc} \text{(AT)} & \text{(C)} & \text{(D)} & \text{(NT)} \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{array} \begin{array}{c} (Z=0) \\ (Z=1) \end{array} \quad \text{and} \quad \mathbf{R}_1 = \begin{array}{c} \begin{array}{cccc} \text{(AT)} & \text{(C)} & \text{(D)} & \text{(NT)} \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \end{array} \begin{array}{c} (Z=0) \\ (Z=1) \end{array}$$

IV Mixture Model

Using this notation, we have the following system for $d \in \mathcal{D}$ and $y \in \mathcal{Y}$

$$\mathbf{R}_d \boldsymbol{\xi}_d(y) = \mathbf{pf}_d(y) \quad \text{and} \quad \boldsymbol{\xi}_d(y) \geq 0 \tag{$\star$}$$

When ($\star$) is underdetermined, $\boldsymbol{\xi}_d(y)$ is generally not point-identified.

Proposition 1 (due to Heckman and Pinto (2018))

For any $\boldsymbol{c} \in \mathbb{R}^{|\mathcal{T}|}$,

$$\boldsymbol{c} \in \text{row-span}(\mathbf{R}_d) \implies \boldsymbol{c}^\top \boldsymbol{\xi}_d \text{ is point-identified },$$

$$\boldsymbol{c} \in \text{row-span}\left(\begin{bmatrix} \mathbf{R}_1 \\ \vdots \\ \mathbf{R}_{N_D} \end{bmatrix} \right) \implies \boldsymbol{c}^\top \boldsymbol{\pi} \text{ is point-identified }.$$

→ Equivalent to Theorem T-2 in Heckman and Pinto (2018), where it is stated in terms of the Moore-Penrose generalized inverse of $\mathbf{R}_d$

Informal Outline of Approach

Step 0: Selection model determines set of possible response-types, $\mathcal{T}$

Step 1: For each $d \in \mathcal{D}$, $\boldsymbol{\xi}_d(y)$ must be a solution to ($\star$) which imposes

1a. restrictions on $\boldsymbol{\pi}$, i.e. distribution of $\boldsymbol{T}$

1b. restrictions on the conditional distribution of $Y_d | \boldsymbol{T}$

Step 2: Get identified set for $\boldsymbol{\pi}$ by combining restrictions from Step 1 over treatments $d \in \mathcal{D}$

- Optional: incorporate additional assumptions on treatment selection
- Reject selection model iff this set is empty (sharp specification test)

Step 3: Define parameter of interest (e.g., ATE)

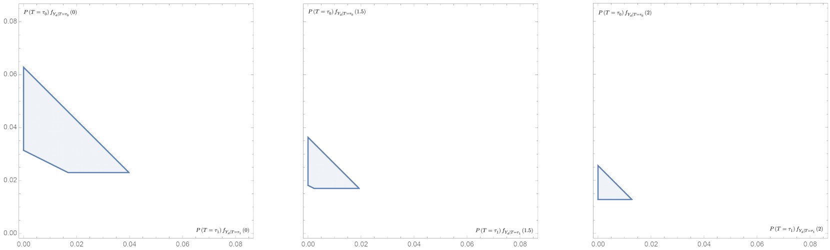
- Possible values for this parameter are those consistent with restrictions from Steps 1 and 2

$\xi_d(y)$ must be a solution to $(\star)$

Fix $d \in \mathcal{D}$. For $y \in \mathcal{Y}$, the set of solutions to $(\star)$ define a convex polytope

$$\mathcal{B}_d\left(\mathbf{pf}_d(y)\right) := \left\{\mathbf{x} \in \mathbb{R}_{\geq 0}^{|\mathcal{T}|} \;\middle|\; \mathbf{R}_d\mathbf{x} = \mathbf{pf}_d(y)\right\} .$$

$\rightarrow$ By $(\star)$, we require that $\xi_d(y) \in \mathcal{B}_d\left(\mathbf{pf}_d(y)\right)$ for each $y \in \mathcal{Y}$.



(a) $\tilde{\mathcal{B}}_d\left(\mathbf{pf}_d(0)\right)$ **(b)** $\tilde{\mathcal{B}}_d\left(\mathbf{pf}_d(1.5)\right)$ **(c)** $\tilde{\mathcal{B}}_d\left(\mathbf{pf}_d(2)\right)$

Figure 6: Hypothetical $\tilde{\mathcal{B}}_d\left(\mathbf{pf}_d(\cdot)\right)$ in a case with 2 "degrees of freedom"

$\xi_d\,(y)$ must be a solution to $(\star)$

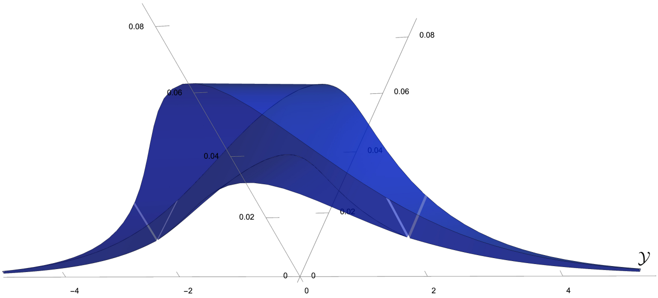


Figure 7: Hypothetical graph of the correspondence $y \mapsto \tilde{\mathcal{B}}_d\left(\mathbf{pf}_d\,(y)\right)$

$\xi_d(y)$ must be a solution to $(\star)$

→ If $\mathbf{x}$ is a vector-valued function such that $\mathbf{x}(y) \in \mathcal{B}_d(\mathbf{pf}_d(y))$ for all $y \in \mathcal{Y}$, then $\mathbf{x}$ is a "candidate" for $\xi_d(y)$.

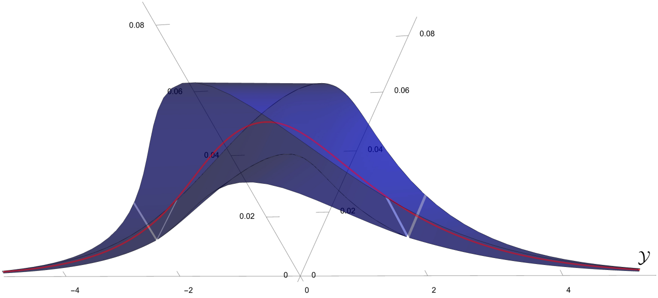
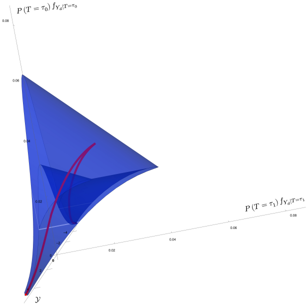


Figure 7: Hypothetical graph of the correspondence $y \mapsto \tilde{\mathcal{B}}_d(\mathbf{pf}_d(y))$

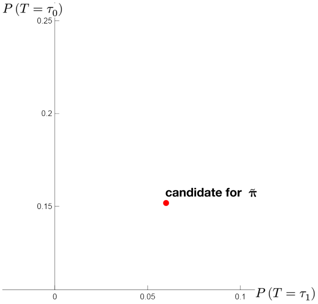
Candidates for π

If $\mathbf{x}$ is a candidate for $\boldsymbol{\xi}_d(y)$, then $\int_{\mathcal{Y}} \mathbf{x}(y)\ d\mu(y)$ is a candidate for $\boldsymbol{\pi}$, the vector of response-type probabilities. Precisely,

$$\mathbf{x} = \boldsymbol{\xi}_d \implies \int_{\mathcal{Y}} \mathbf{x}(y)\ d\mu(y) = \boldsymbol{\pi}$$



(a) $\mathbf{x}(y) \in \mathcal{B}_d(\mathbf{pf}_d(y))\ \forall y \in \mathcal{Y}$



(b) $\int_{\mathcal{Y}} \mathbf{x}(y)\ d\mu(y)$

Candidates for $\boldsymbol{\pi}$

If $\mathbf{x}$ is a candidate for $\boldsymbol{\xi}_d(y)$, then $\int_{\mathcal{Y}} \mathbf{x}(y)\ d\mu(y)$ is a candidate for $\boldsymbol{\pi}$, the vector of response-type probabilities. Precisely,

$$\mathbf{x} = \boldsymbol{\xi}_d \implies \int_{\mathcal{Y}} \mathbf{x}(y)\ d\mu(y) = \boldsymbol{\pi}$$

Define $\Pi_{\mathcal{T}|d}$ as the set of such candidates for $\boldsymbol{\pi}$:

$$\begin{aligned}\Pi_{\mathcal{T}|d} &:= \left\{ \int_{\mathcal{Y}} \mathbf{x}(y)\ d\mu(y) \;\middle|\; \begin{array}{l} \text{measurable } \mathbf{x} : \mathcal{Y} \to \mathbb{R}_{\geq 0}^{|\mathbf{T}|} \\ \mathbf{R}_d \mathbf{x}(y) = \mathbf{pf}_d(y),\ \forall y \in \mathcal{Y} \end{array} \right\} \\ &= \left\{ \int_{\mathcal{Y}} \mathbf{x}(y)\ d\mu(y) \;\middle|\; \begin{array}{l} \text{measurable } \mathbf{x} : \mathcal{Y} \to \mathbb{R}^{|\mathbf{T}|} \\ \mathbf{x}(y) \in \mathcal{B}_d(\mathbf{pf}_d(y)),\ \forall y \in \mathcal{Y} \end{array} \right\}.\end{aligned}$$

→ $\Pi_{\mathcal{T}|d}$ is a convex polytope, and its boundary can be obtained by integrating along the boundary of $\mathcal{B}_d(\mathbf{pf}_d(y))$

Candidates for $\boldsymbol{\pi}$

Define $\Pi_{\mathcal{T}|d}$ as the set of such candidates for $\boldsymbol{\pi}$:

$$\Pi_{\mathcal{T}|d} := \left\{ \int_{\mathcal{Y}} \mathbf{x}(y)\, d\mu\,(y) \;\middle|\; \begin{array}{l} \text{measurable } \mathbf{x} : \mathcal{Y} \to \mathbb{R}^{|\mathbf{T}|} \\ \mathbf{x}\,(y) \in \mathcal{B}_d\left(\mathbf{pf}_d\,(y)\right),\ \forall y \in \mathcal{Y} \end{array} \right\}.$$

Comments

- We must have $\boldsymbol{\pi} \in \Pi_{\mathcal{T}|d}$ for every $d \in \mathcal{D}$.
- $\Pi_{\mathcal{T}|d} = \emptyset$ iff $\mathcal{B}_d\left(\mathbf{pf}_d\,(y)\right) = \emptyset$ for some $y \in \mathcal{Y}$.
- $\Pi_{\mathcal{T}|d}$ is the *Minkowski integral* of the map $y \mapsto \mathcal{B}_d\left(\mathbf{pf}_d\,(y)\right)$
  - Continuous analogue to Minkowski sum (Billera and Sturmfels, 1992)
  - Close connection to *Aumann expectation* from theory of random sets (Artstein and Vitale, 1975): both are adaptations of Aumann (1965)'s integral for set-valued measures

Candidates for π: Support Function

For any non-empty closed convex set $C \subseteq \mathbb{R}^n$, let its ***support function*** be defined pointwise, for $\boldsymbol{c} \in \mathbb{R}^n$, as

$$\nu\left(\boldsymbol{c}; C\right) := \sup_{\boldsymbol{x} \in C} \boldsymbol{c}^\top \boldsymbol{x} \ .$$

$\rightarrow$ Every non-empty closed convex set is uniquely determined by its support function.

Lemma 1

Suppose $\mathcal{T} \subseteq \mathcal{D}^{\mathcal{Z}}$ is such that $\mathcal{B}_d\left(\mathbf{pf}_d\left(y\right)\right) \neq \emptyset$ for all $y \in \mathcal{Y}$. Then $\Pi_{\mathcal{T}|d}$ is a non-empty convex set such that, for all $\boldsymbol{c} \in \mathbb{R}^{|\mathcal{T}|}$

$$\nu\left(\boldsymbol{c}; \Pi_{\mathcal{T}|d}\right) = \int_{\mathcal{Y}} \nu\left(\boldsymbol{c};\ \mathcal{B}_d\left(\mathbf{pf}_d\left(y\right)\right)\right)\ d\mu\left(y\right) \ .$$

Further, $\Pi_{\mathcal{T}|d}$ is a convex polytope.

Identified Set for Response-Type Probabilities

Let $\mathcal{Q}$ be the set of joint distributions $((Y_d : d \in \mathcal{D}), (D_z : z \in \mathcal{Z}), Z)$ that are consistent with the model and the observed data. For $\mathcal{T} \subseteq \mathcal{D}^{\mathcal{Z}}$, define

$$\mathcal{Q}_{\mathcal{T}} := \left\{ Q \in \mathcal{Q} \,\middle|\, \operatorname{supp}_Q(\boldsymbol{T}) = \mathcal{T} \right\} ,$$

and define the identified set for $\boldsymbol{\pi}$, under the assumption $\operatorname{supp}(\boldsymbol{T}) = \mathcal{T}$,

$$\Pi_{\mathcal{T}} := \left\{ \Big(Q(\boldsymbol{T} = \boldsymbol{\tau}) : \boldsymbol{\tau} \in \mathcal{T} \Big) \,\middle|\, Q \in \mathcal{Q}_{\mathcal{T}} \right\} .$$

Theorem 1 (Identified Set for Response-Type Probabilities)

For any $\mathcal{T} \subseteq \mathcal{D}^{\mathcal{Z}}$, $\Pi_{\mathcal{T}}$ is a compact convex polytope and

$$\Pi_{\mathcal{T}} = \bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}|d} .$$

Moreover, $\mathcal{Q}_{\mathcal{T}} \neq \emptyset$ if and only if $\bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}|d} \neq \emptyset$.

Some Implications of Theorem 1

Denote by $\left\{\mathcal{Y} \to \mathbb{R}_{\geq 0}^{|\mathcal{T}|}\right\}$ the set of measurable functions $\mathbf{x} : \mathcal{Y} \to \mathbb{R}_{\geq 0}^{|\mathcal{T}|}$.

Corollary 1 (Identified Set for Marginal Outcome Distributions)

Fix $\mathcal{T} \subseteq \mathcal{D}^{\mathcal{Z}}$ and suppose $\operatorname{supp}(\boldsymbol{T}) = \mathcal{T}$. Then, for any $d \in \mathcal{D}$, the identified set for $\boldsymbol{\xi}_d$ is

$$\Xi_{\mathcal{T}}(d) := \left\{ \mathbf{x} \in \left\{\mathcal{Y} \to \mathbb{R}_{\geq 0}^{|\mathcal{T}|}\right\} \;\middle|\; \begin{array}{l} \mathbf{R}_d \mathbf{x}(y) = \mathbf{pf}_d(y), \ \forall y \in \mathcal{Y}; \\ \int_{\mathcal{Y}} \mathbf{x}(y) \ d\mu(y) \in \Pi_{\mathcal{T}} \end{array} \right\}.$$

Further, the joint identification region for $(\boldsymbol{\xi}_d : d \in \mathcal{D})$ is

$$\left\{ (\mathbf{x}_d : d \in \mathcal{D}) \;\middle|\; \begin{array}{l} \forall d \in \mathcal{D}, \mathbf{x}_d \in \Xi_{\mathcal{T}}(d); \\ \exists \boldsymbol{p} \in \Pi_{\mathcal{T}} \text{ s.t. } \forall d \in \mathcal{D}, \int_{\mathcal{Y}} \mathbf{x}_d(y) \ d\mu(y) = \boldsymbol{p} \end{array} \right\}$$

Comments

- Jointly tests $\text{supp}(\boldsymbol{T}) = \mathcal{T}$ and random assignment of Z
    - Assume potential outcomes exist, set $\mathcal{T} = \mathcal{D}^{|\mathcal{Z}|}$ in Theorem 1
      $\implies$ sharp test for random assignment of Z
- Theorem 1 requires (i) $\Pi_{\mathcal{T}|d} \neq \emptyset$ for each $d \in \mathcal{D}$, and (ii) $\bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}|d} \neq \emptyset$
- $\Pi_{\mathcal{T}|d} \neq \emptyset$ iff $\mathbf{pf}_d(y) \in \text{cone}(\mathbf{R}_d)$ for all $y \in \mathcal{Y}$
    - $\text{cone}(\mathbf{R}_d)$ is the polyhedral cone generated by positive linear combinations of the columns of $\mathbf{R}_d$
    - (i) is automatically satisfied in some cases; e.g., $\mathcal{T} = \mathcal{D}^{|\mathcal{Z}|}$
    - Possible to represent a joint test for (i) and (ii) as a test of conditional moment inequalities

Remaining content of the paper

1. Identification Remaining identification results, including under some assumptions on outcomes
2. Analytic bounds: Automated derivation of analytic expressions for “point-wise” bounds
3. Inference: Support function based test statistic for inclusion tests
4. Computation: Primal-dual method for computing support function
    - feasible for large problems, exact solution in finitely many steps
    - also useful for fast approximation with deterministic error bounds
5. Theory: Formalization of high-level claims made in the presentation

Conclusion

Summary

Framework for partial identification in IV treatment selection models

- multivalued treatments, discrete instruments
- sharp under random assignment of instrument
- tractable, not “black box”: analytic bounds
- sharp specification tests

Current and Future Work

- Formalizing inference and estimation
- Simulations and empirical applications
- Developing a Stata/R/Python package
- Additional guidance on when bounds are likely to be informative

Thank you!

Basic Example ($Z \in \{0,1\}$, $D \in \{0,1\}$, no selection restrictions)

For $d = 1$, ($\star$) implies

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \boldsymbol{\xi}_1(y) = \mathbf{pf}_1(y)$$

$$\Longrightarrow \begin{bmatrix} 1 \\ 1 \end{bmatrix} P(\boldsymbol{T} = AT) f_{Y_1|\boldsymbol{T}=AT}(y) \leq \begin{bmatrix} P(D=1|Z=0) f_{Y|D=1,Z=0}(y) \\ P(D=1|Z=1) f_{Y|D=1,Z=1}(y) \end{bmatrix}$$

The support function for the 1-dimensional polytope (i.e. interval) defined by the non-negative solutions to the above system is simply the map

$$c \mapsto \begin{cases} c\left(\min_{z\in\{0,1\}} \left\{P(D=1|Z=z) f_{Y|D=1,Z=z}(y)\right\}\right) & \text{if } c \geq 0 \\ 0 & \text{if } c < 0 \end{cases}$$

Basic Example ($Z \in \{0,1\}$, $D \in \{0,1\}$, no selection restrictions)

$\Pi_{\mathcal{T}|1}$ consists of all $\boldsymbol{\pi} \geq 0$ such that

$$\boldsymbol{\pi}_{(AT)} \leq \int_{\mathcal{Y}} \min_{z\in\{0,1\}} \left\{P\left(D=1|Z=z\right) f_{Y|D=1,Z=z}\left(y\right)\right\} \, d\mu\left(y\right),$$

$$\boldsymbol{\pi}_{(AT)} + \boldsymbol{\pi}_{(D)} = P\left(D=1|Z=0\right) = \int_{\mathcal{Y}} P\left(D=1|Z=0\right) f_{Y|D=1,Z=0}\left(y\right) \, d\mu\left(y\right),$$

$$\boldsymbol{\pi}_{(AT)} + \boldsymbol{\pi}_{(C)} = P\left(D=1|Z=1\right) = \int_{\mathcal{Y}} P\left(D=1|Z=1\right) f_{Y|D=1,Z=1}\left(y\right) \, d\mu\left(y\right).$$

Similarly, $\Pi_{\mathcal{T}|0}$ consists of all $\boldsymbol{\pi} \geq 0$ such that

$$\boldsymbol{\pi}_{(NT)} \leq \int_{\mathcal{Y}} \min_{z\in\{0,1\}} \left\{P\left(D=0|Z=z\right) f_{Y|D=0,Z=z}\left(y\right)\right\} \, d\mu\left(y\right),$$

$$\boldsymbol{\pi}_{(C)} + \boldsymbol{\pi}_{(NT)} = P\left(D=0|Z=0\right),$$

$$\boldsymbol{\pi}_{(D)} + \boldsymbol{\pi}_{(NT)} = P\left(D=0|Z=1\right).$$

$\Pi_{\mathcal{T}}$ consists of all $\boldsymbol{\pi} \geq 0$ that satisfy the 6 conditions above.

Back to Theorem 1

MTO Example: Sharp Bounds on $\pi_{[(\cdot\cdot M)]}$

For the MTO example considered earlier, the following bounds are sharp

$$\max\left\{\int_{\mathcal{Y}} \max\left\{0, \mathbf{pf}_{M|Z=(1,1)}(y) - \mathbf{pf}_{M|Z=(0,0)}(y)\right\} d\mu(y),\right.$$

$$\left.\mathbf{p}_H[(0,1)] - \mathbf{p}_H[(1,1)] \int_{\mathcal{Y}} \max\left\{0, \mathbf{pf}_{L|Z=(0,0)}(y) - \mathbf{pf}_{L|Z=(1,1)}(y)\right\} d\mu(y)\right\}$$

$$\leq \boldsymbol{\pi}_{(HHM)} + \boldsymbol{\pi}_{(HLM)} + \boldsymbol{\pi}_{(LLM)} \leq$$

$$\min\left\{\mathbf{p}_M[(1,1)] - \mathbf{p}_M[(0,1)],\right.$$

$$\left.\mathbf{p}_M[(1,1)] - \mathbf{p}_M[(0,0)] + \int_{\mathcal{Y}} \min\left\{\mathbf{pf}_{L|Z=(1,1)}(y), \mathbf{pf}_{L|Z=(0,1)}(y) - \mathbf{pf}_{L|Z=(0,0)}(y)\right\} d\mu(y)\right\}$$

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